

LESSON 2-5 • HOMEWORK PACKET • LEVEL 1 • SHEET 1 OF 2**Range and IQR**

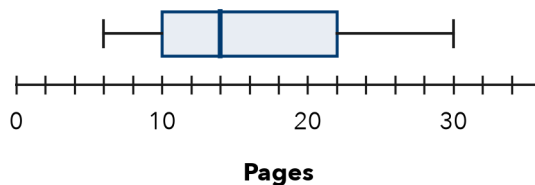
Maryland Standard 6.SP.B.5c • Learning target – I can find the range and the interquartile range of a data set.

REMEMBER

- **Range** = maximum – minimum. It is the spread of all the data.
- **IQR** (interquartile range) = $Q3 - Q1$. It is the spread of the middle half.
- One extreme value changes the range a lot but the IQR very little.

WORKED EXAMPLE

Find the range and IQR of 3, 7, 8, 10, 12, 15, 20.

Step 1 Range: $20 - 3 = 17$ **Step 2** Median 10. $Q1 = 7$ (of 3, 7, 8). $Q3 = 15$ (of 12, 15, 20).**Step 3** IQR: $15 - 7 = 8$ **Answer Range 17, IQR 8****WORD BANK****range** – biggest number – smallest number**IQR** – $Q3 - Q1$ **$Q1, Q3$** – the middles of the lower and upper halves**minimum** – the smallest value**PRACTICE***show your work • label units***1.** Find each measure.**a.** Range of 14, 6, 21, 9: _____**b.** Range of 52, 38, 47, 60: _____**c.** IQR when $Q1 = 18, Q3 = 31$: _____**d.** IQR when $Q1 = 40, Q3 = 56$: _____**e.** Range of 75, 90, 68, 83: _____**f.** IQR when $Q1 = 7, Q3 = 25$: _____**2. Pages Read.** The box plot shows pages read by students in one week.**Step 1** Read the ends of the whiskers: max – min = range.**Step 2** Read the box edges: $Q3 - Q1 = \text{IQR}$.**a.** Range: _____ pages**b.** IQR: _____ pages

- 3. Jump Rope.** Seven students jumped rope 35, 52, 41, 60, 28, 47, and 55 times. Find the range and the IQR.

Step 1 Put the 7 numbers in order.

Step 2 Range = last – first.

Step 3 Find Q1 and Q3, then subtract.

a. Range: _____

b. IQR: _____

Show your work.

- 4. Work Backward.** A data set has a **minimum** of 14 and a **range** of 29. Its Q1 is 20 and its IQR is 11.

Step 1 Maximum = minimum + range.

Step 2 Q3 = Q1 + IQR.

a. Maximum: _____

b. Q3: _____

- 5. Find the Error.** Mateo says the IQR is always half of the range. Use the Jump Rope data to explain why he is wrong.

Frame Half of the range is ____, but the IQR is ____, so Mateo is ____.

CHECK YOURSELF

☐ I can do this on my own

☐ I need a little help

☐ I have a question about #____

AT HOME Ask family members their ages. Find the range of the ages.

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Comparing Variability

Maryland Standard 6.SP.B.5c • Learning target – I can compare the variability of two data sets using range and IQR.

REMEMBER

- More range or IQR = **more variability**.
- Less variability = more **consistent**. Compare range with range, IQR with IQR.
- With an extreme value, use the **IQR**.

WORKED EXAMPLE

Team P: IQR 6 points. Team Q: IQR 15 points. Which is more consistent?

Step 1 $6 < 15$, so Team P's middle half is closer.

Answer Team P

WORD BANK

consistent – close together

variability – how spread out the data are

range – $\text{max} - \text{min}$

IQR – $Q3 - Q1$

PRACTICE

show your work • label units

1. Find each IQR.

a. Set P: $Q1 = 30$, $Q3 = 42$ _____

b. Set Q: $Q1 = 33$, $Q3 = 39$ _____

2. **Basketball.** Points per game: Player 1: 8, 10, 11, 12, 13, 14, 16. Player 2: 2, 6, 9, 12, 15, 18, 22.

Step 1 Range = greatest – least for each player.

Step 2 IQR = $Q3 - Q1$ for each player.

Step 3 Smaller numbers mean more consistent.

	Range	IQR
Player 1		
Player 2		

More consistent player _____

3. **Spelling Scores.** Group A scored 70, 72, 75, 78, 80. Group B scored 55, 65, 75, 85, 95.

Ranges: A _____ B _____ More consistent group: _____

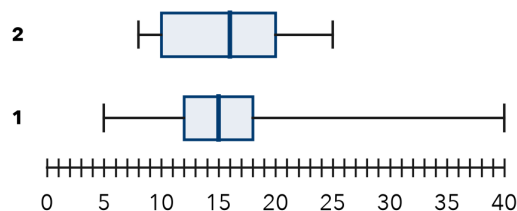
4. **Bus Routes.** The box plots show ride times in minutes.

Step 1 Read the whisker ends for each range.

Step 2 Read the box edges for each IQR.

a. Ranges: 1 _____ 2 _____

b. IQRs: 1 _____ 2 _____



5. Jada says Route 2 is more consistent. Kwame says Route 1 is. Who is right? Why?

Frame ____ is right because Route 1's range is big only because ____, and its IQR is ____.

TEST PRACTICE

choose the best answer

6. Class A's scores have a range of 12. Class B's have a range of 30. Whose scores vary less?

A Class A

B Class B

C Both the same

D Cannot tell

7. What is the IQR of 4, 6, 7, 9, 10, 12, 15?

A 6

B 11

C 4.5

D 9

CHECK YOURSELF

☐ I can do this on my own

☐ I need a little help

☐ I have a question about #____